Confused and Dehydrated Hyper & Hypo-calcaemia

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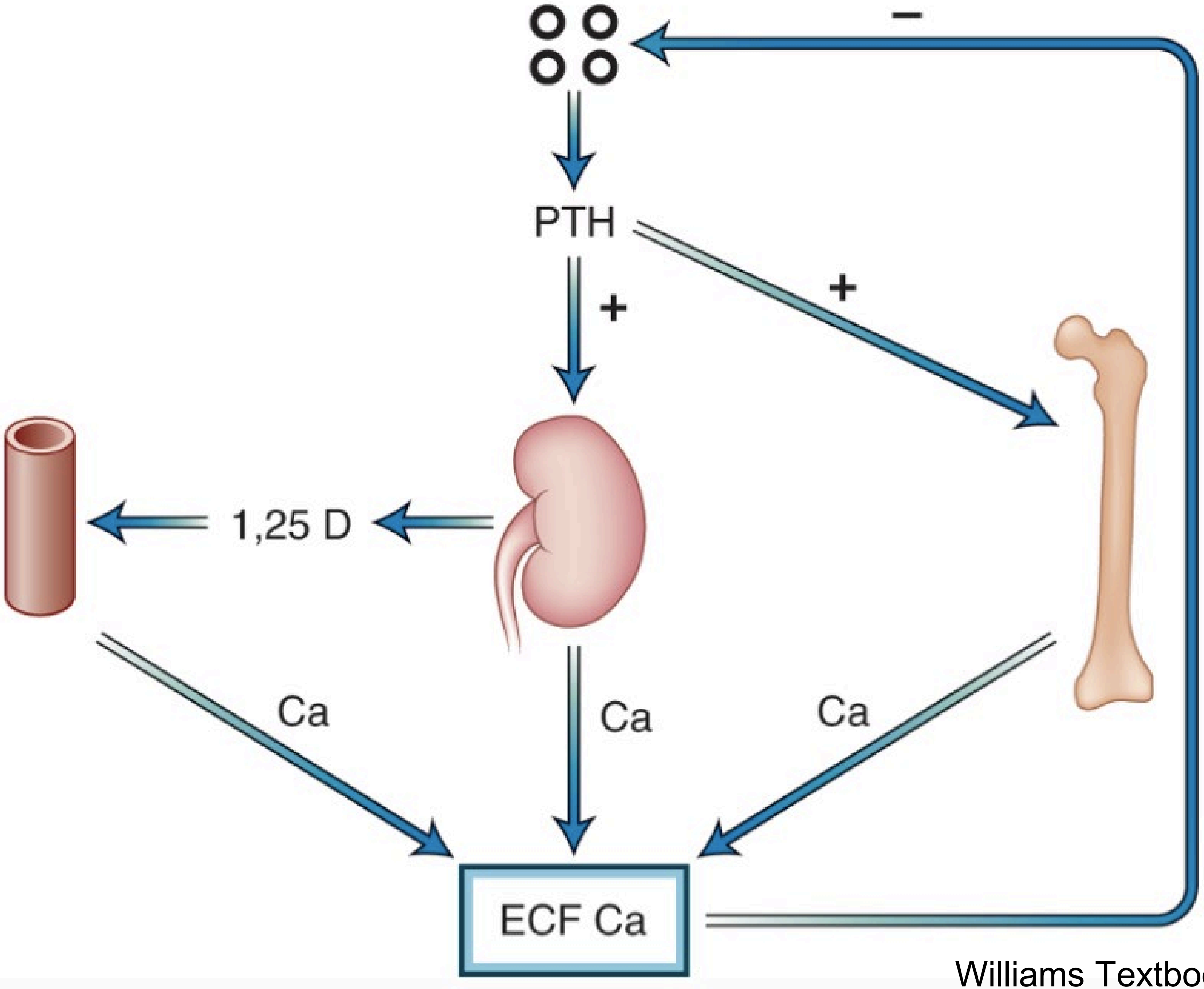
Case Vignette

- Madam Chan
- F/52
- Incidental health check found hypercalcaemia
- Asymptomatic
- Blood tests showed high serum calcium, low phosphate with serum parathyroid hormone level at high normal range

Serum Calcium Control



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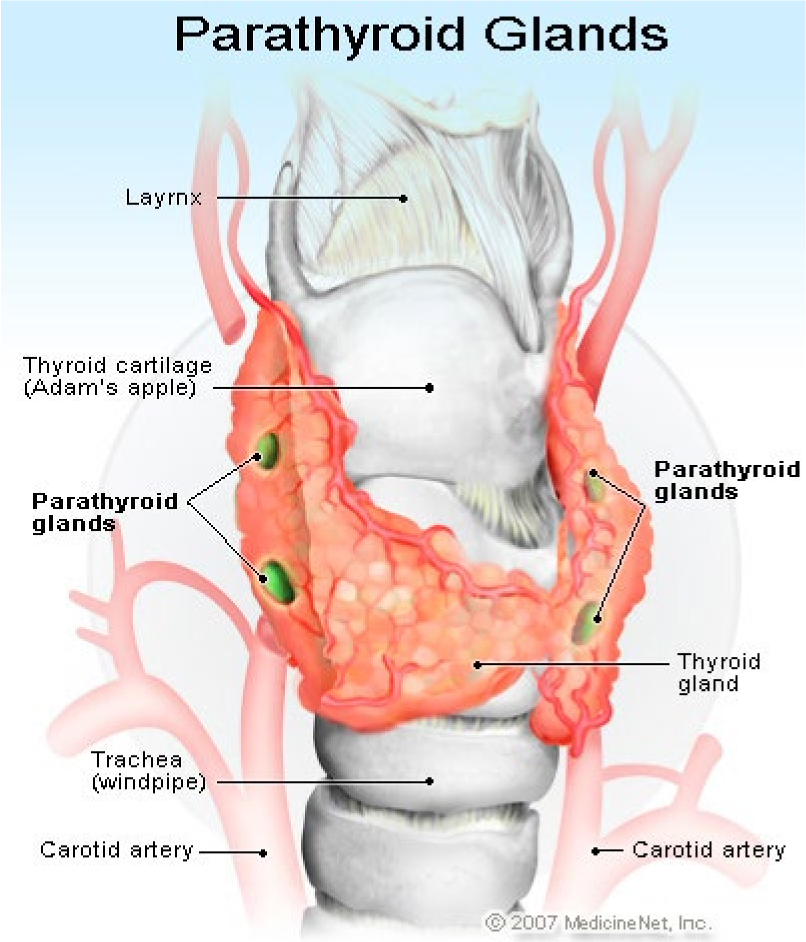
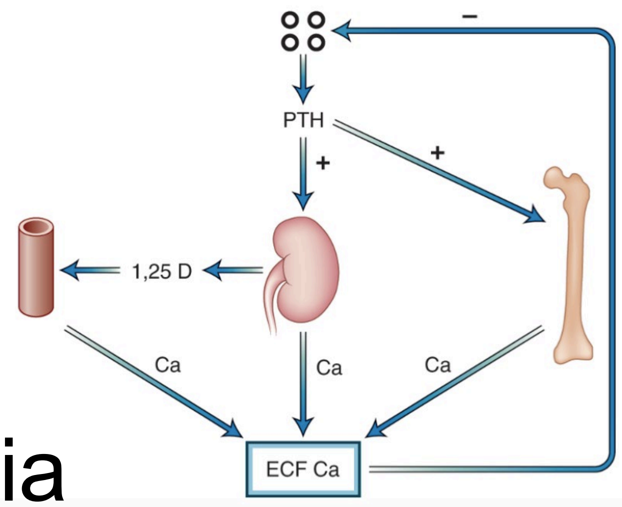
Parathyroid Glands



- There are usually four parathyroid glands located behind the left and right lobes of the thyroid
- The major function of the parathyroid glands is to maintain the body's calcium and phosphate levels within a very narrow range, so that the neuromuscular systems can function properly.

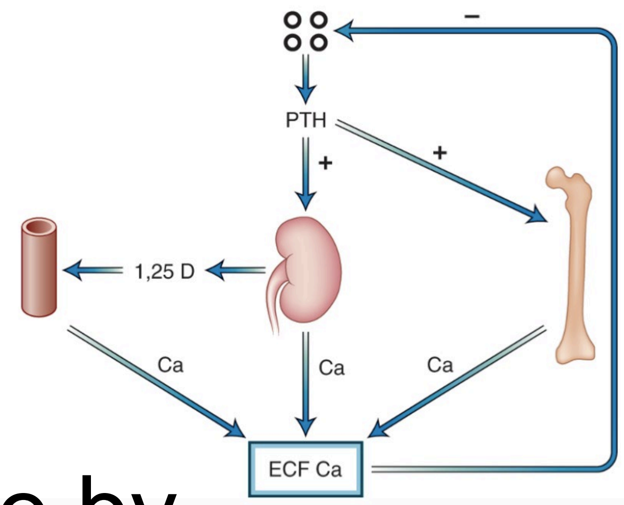
Parathyroid Hormone



<u>Physiological effects</u>

- Increase calcium resorption from bone via osteoblasts & osteoclasts
- Increase reabsorption of calcium from distal convoluted tubules
- Increase uptake of calcium from intestine via its effect on the formation of $1,25(OH)_2D_3$
- Decrease phosphate reabsorption from proximal convoluted tubules

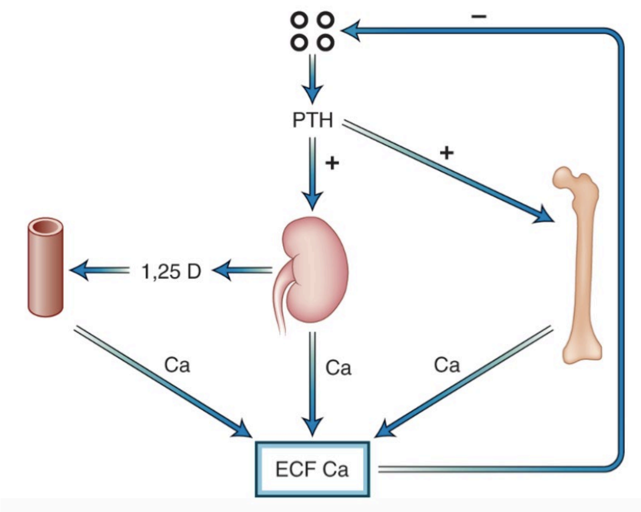
Vitamin D



Physiological effects

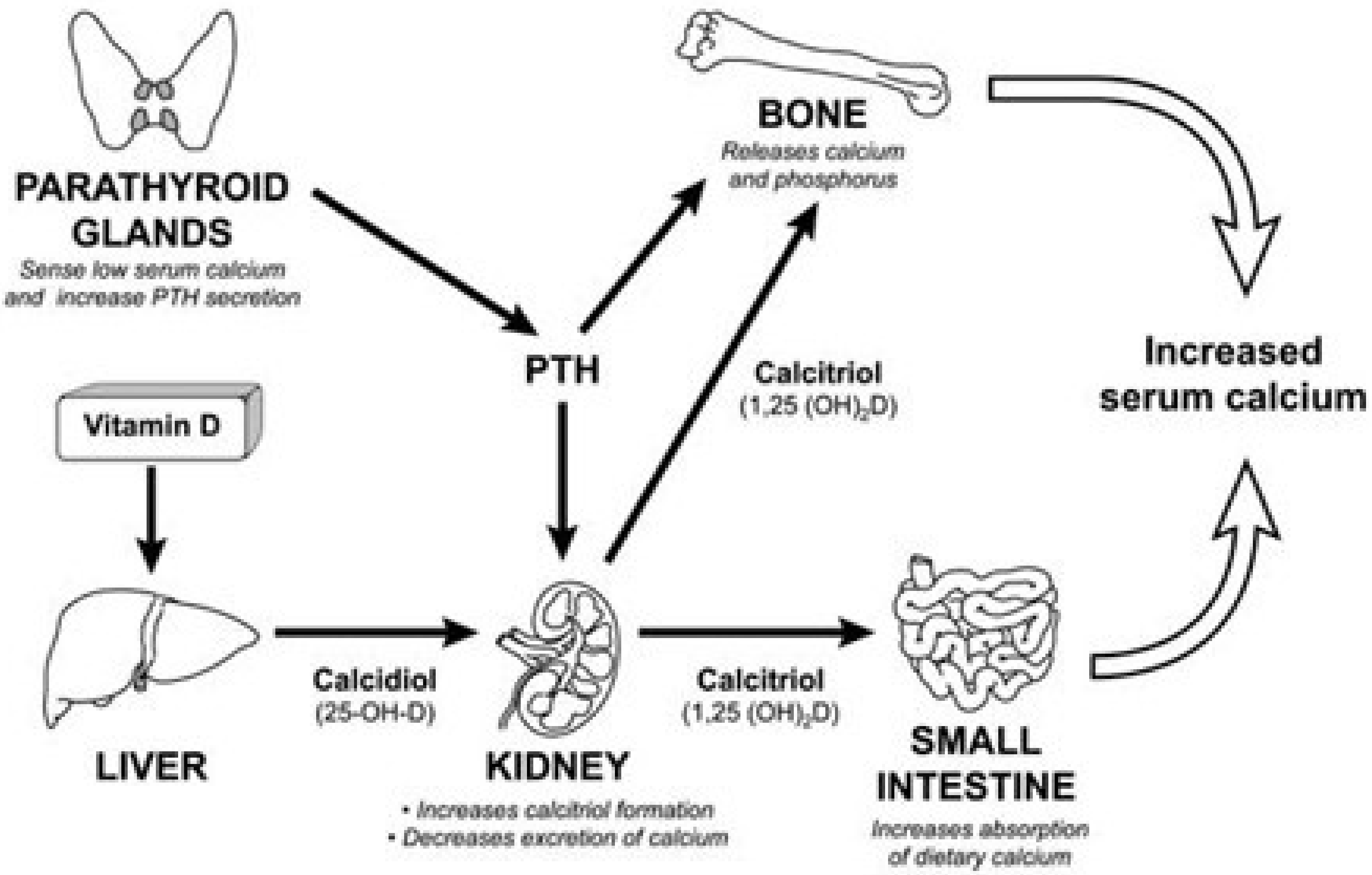
- Increase calcium uptake in intestine by increasing calcium binding protein
- Increase intestinal absorption of phosphate
- Increase calcium resorption from bone (if in high doses)

Calcitonin



<u>Physiological effects</u>

- Decrease osteoclast mediated bone resorption
- Decrease reabsorption of calcium from kidney



PARATHYROID
GLANDS
Sense low serum calcium
and increase PTH secretion
BONE
Releases calcium
and phosphorus
PTH
Calcitriol
(1,25 (OH)$_2$D)
Increased
serum calcium
Vitamin D
LIVER
Calcidiol
(25-OH-D)
KIDNEY
• Increases calcitriol formation
• Decreases excretion of calcium
Calcitriol
(1,25 (OH)$_2$D)
SMALL
INTESTINE
Increases absorption
of dietary calcium

Calcium Homeostasis - Other Factors

- Albumin (50% of plasma calcium)
- Phosphate (affects ionic balance of calcium)
- pH (affects ionic calcium level)

Adjusted or Corrected Calcium

- With hypoalbuminaemia, total calcium may be low yet the ionized calcium is normal
- Corrected Ca (mmol/L)

  = Total Ca (mmol/L) + [0.02 x (40 – albumin in g/L)]
- This is an approximation
- Measurement of ionized calcium is better

Example

Ca (total)		2.20 mmol/L
Albumin		25 g/L
Corrected Ca	=	2.20 + [0.02 x (40-25)]
	=	2.20 + 0.3
	=	2.50 mmol/l

Pseudohypocalcaemia

Ionized calcium will be normal

Nephrotic syndrome or cirrhosis

Causes of Hypercalcaemia

Causes of Hypercalcaemia

1. Hyperparathyroidism
   - Primary (Common)
   - Tertiary

2. Hypercalcaemia of malignancy

Causes of Hypercalcaemia

3. Excessive administration of vitamin D and or calcium
   - Vitamin D intoxication
   - Milk alkali syndrome
4. Increased sensitivity to vitamin D
   - Tuberculosis, sarcoidosis
5. Hypocalciuric hypercalcaemia
   - Thiazide diuretics
   - Familial (mutation of calcium sensing receptor CaSR)
6. Uncommon causes
   - Adrenal insufficiency
   - Hyperthyroidism

Causes of Hypercalcaemia

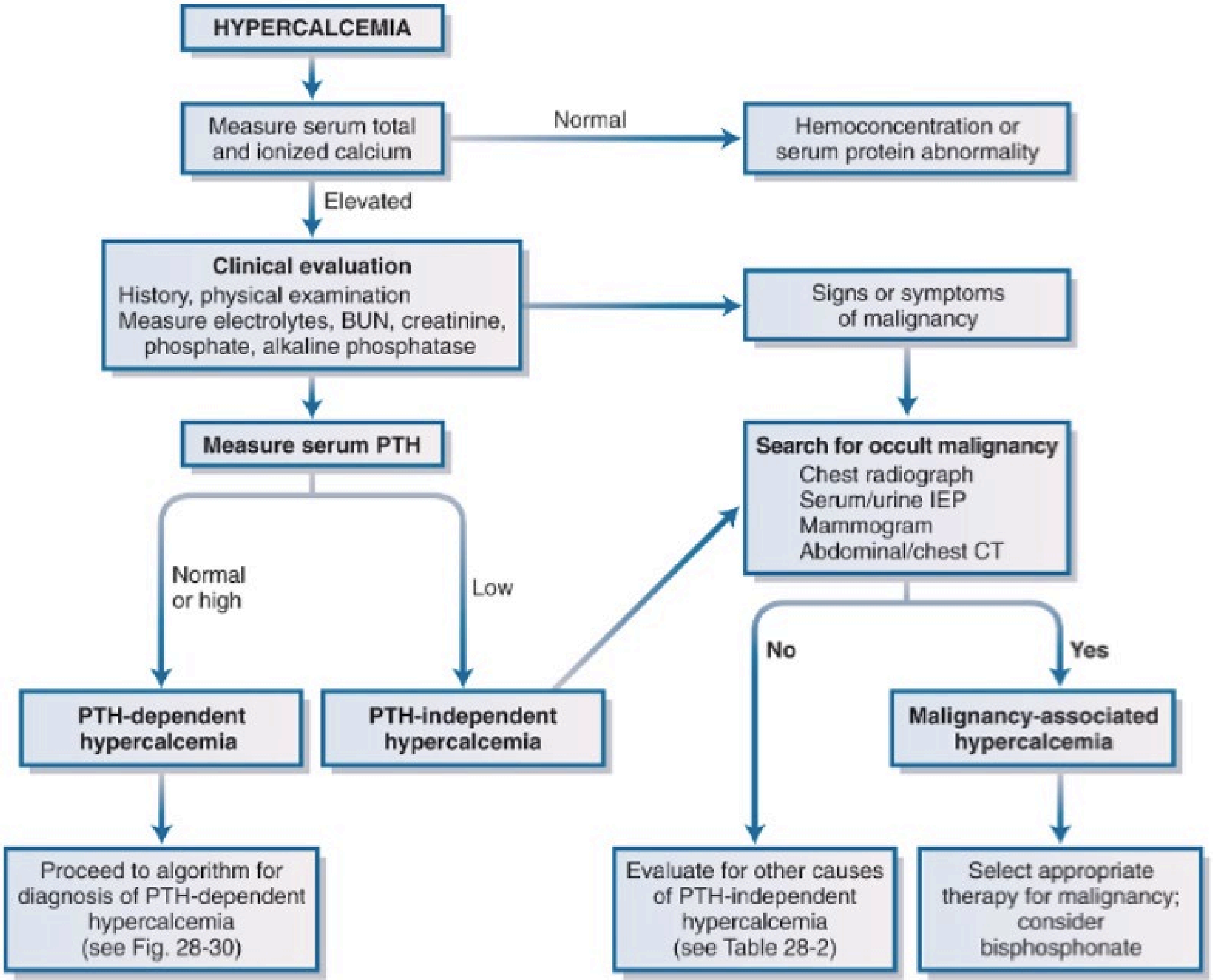
Adrenal insufficiency and hypercalcaemia

Possible mechanisms

- Hypovolaemia → Reduction in GFR → Reduce calcium filtered → Increase calcium renal reabsorption

- Increase 1-alpha-ydroxylase activity → Increase intestinal absorption of calcium

Bhatti RS BMJ Case Rep 2012
Minisola S et al. BMJ 2015



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Management of
Severe Hypercalcaemia

Patient Scenario

Comment	Below			
Na	152	H	136 - 148	[illegible]
K	3.9		3.6 - 5.0	[illegible]
Chloride	104		100 - 109	[illegible]
Urea	20.5	H	3.6 - 7.8	[illegible]
Creatinine	190	H	67 - 109	[illegible]
R Glucose	60.3	H	2hr pp < 7.8	[illegible]
Calcium	3.10	H	2.11 - 2.55	[illegible]
Phosphate	1.92	H	0.88 - 1.45	[illegible]
Total Protein	94	H	68 - 84	g/L
Albumin	49		39 - 50	g/L
Globulin	45	H	24 - 37	g/L
Total Bili	9		4 - 23	umol/L
ALP	196	H	42 - 110	U/L
ALT	49		8 - 58	U/L
AST	34		15 - 38	U/L
CK	1468	H	65 - 355	U/L
Troponin I	0.05		See Below	ng/mL
Amylase	983	H	25 - 124	U/L

Comment:
14C1111202 Result Telephoned.

Classification

Asymptomatic or mildly symptomatic hypercalcaemia
- Serum Ca < 3mmol/L

Moderate hypercalcaemia
- Serum Ca 3.0 - 3.5 mmol/L

Severe hypercalcaemia
- Serum Ca > 3.5 mmol/L

Asymptomatic or
Mildly Symptomatic Hypercalcaemia

- Avoid factors that could aggravate hypercalcaemia
- Maintain adequate hydration
- Avoid high calcium diet (>1000mg/day)

Principles for Management of moderate and severe hypercalcaemia

1. Rapid control of calcium levels esp. for moderate and severe hypercalcaemia

2. Early diagnosis of the cause

Fluid Replacement

- Volume expansion with normal saline
- Excretion of calcium achieved by inhibition of sodium reabsorption in proximal tubule and loop of Henle
- Rate of saline infusion depends on other factors, including age, comorbid conditions (heart failure, CKD etc)
- Monitor electrolytes and fluid balance
- Loop diuretics (eg Frusemide) *only* if patients develop oedema

IV Bisphosphonates

- Nonhydrolyzable analogs of inorganic pyrophosphate
- Adsorb to the surface of bone hydroxyapatite and interfere with osteoclast-mediated bone resorption
- Useful agents in moderate or severe hypercalcaemia
- Intravenous pamidronate or zoledronic acid (Zometa)
- Serum calcium might begin to fall in 1-2 days, but maximum effect occurs in 2 to 4 days
- Repeat dosing after a minimum of 7 days
- Contraindicated in patients with eGFR <35ml/min/1.73m^2
- Side effects: flu-like symptoms, renal impairment, osteonecrosis of the jaw (prolonged use) and atypical fractures (prolonged use)

Calcitonin

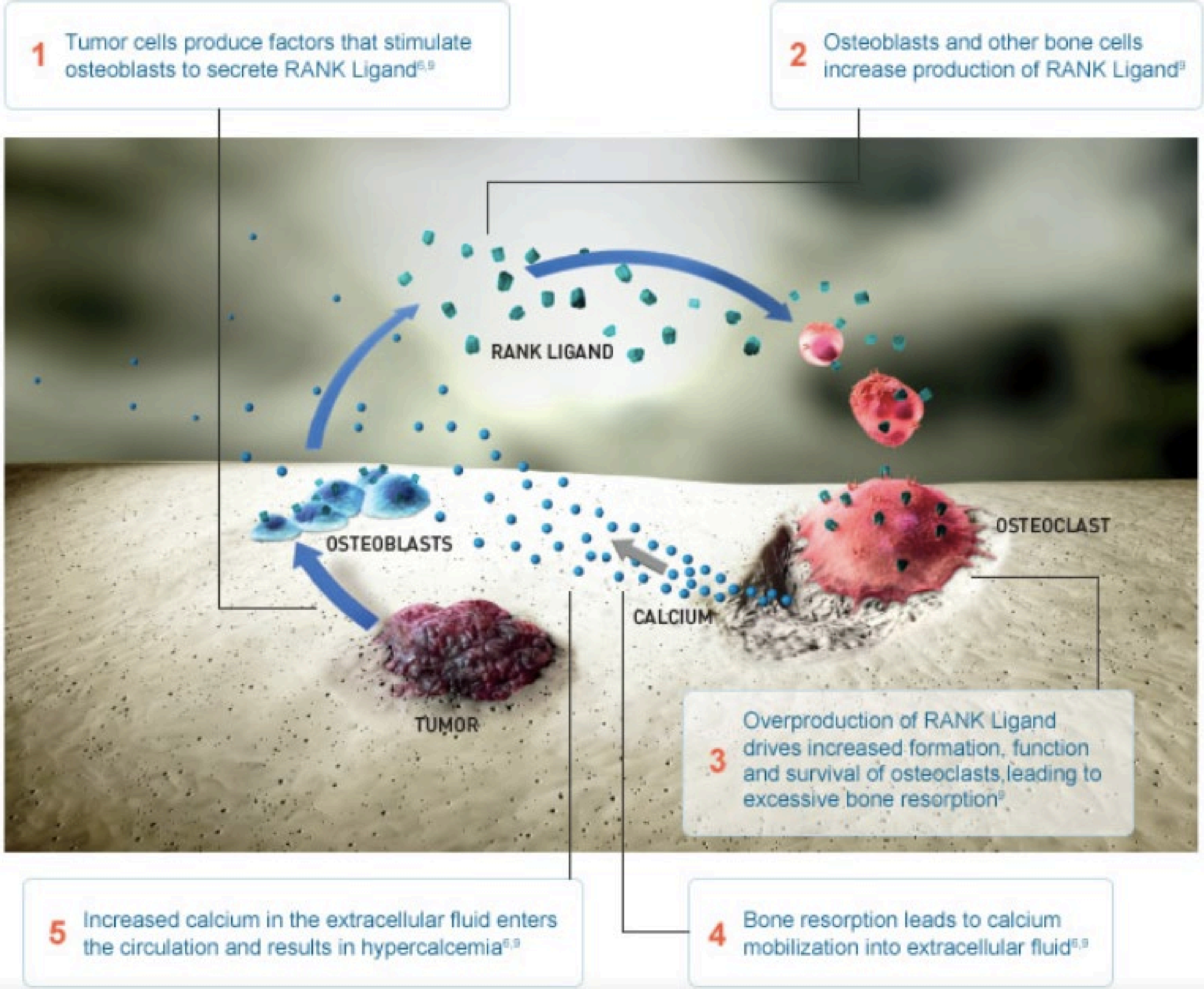
- Increase renal calcium excretion
- Decrease bone resorption
- Relatively weak agent but works rapidly within hours
- Given subcutaneously every 12 hours (salmon calcitonin 4IU/kg); nasal sprays not efficacious for treatment of hypercalcaemia
- Tachyphylaxis is a limitation
- Side effects: Nausea, hypersensitivity reaction (rare)

Glucocorticoids

- Indications:
  - Excessive administration or ingestion of vitamin D
  - Endogenous overproduction of calcitriol (e.g chronic granulomatous diseases, lymphoma)
- Decrease calcitriol production by activated mononuclear cells
- E.g Prednisolone 20-40mg/day

Monoclonal Antibodies against Receptor Activator of Nuclear Factor Kappa B Ligand (RANKL)

- Indications:
  - Refractory hypercalcaemia despite treatment with IV bisphosphonates
  - Contraindicated for IV bisphosphonates due to severe renal impairment
  - Useful agents in hypercalcaemia of malignancy with persistent hypercalcaemia
- Denosumab not excreted through kidneys, hence could be given in patients with CKD
- Could lower serum calcium in 2-4 days
- Ensure vitamin D repleted before administration of denosumab



http://www.xgeva.com/

Dialysis

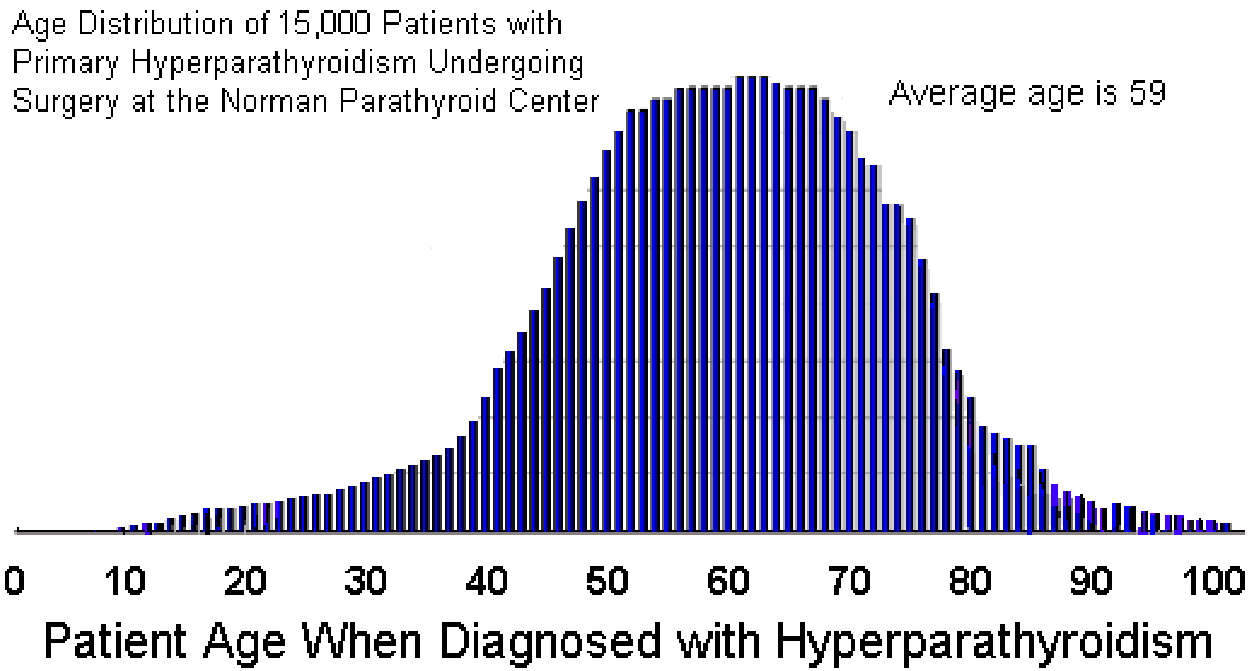
- Reserved for very severe forms of hypercalcaemia (eg. Serum Ca >4.5 mmol/L)
- Refractory severe hypercalcaemia complicated by renal failure which renders other forms of therapy ineffective or contraindicated

Primary Hyperparathyroidism

Primary Hyperparathyroidism

- Most common cause of hypercalcemia
- Prevalence ~1-2/1000
- Peaks in 6th – 7th decade
- F:M ~2-3:1
- Causes

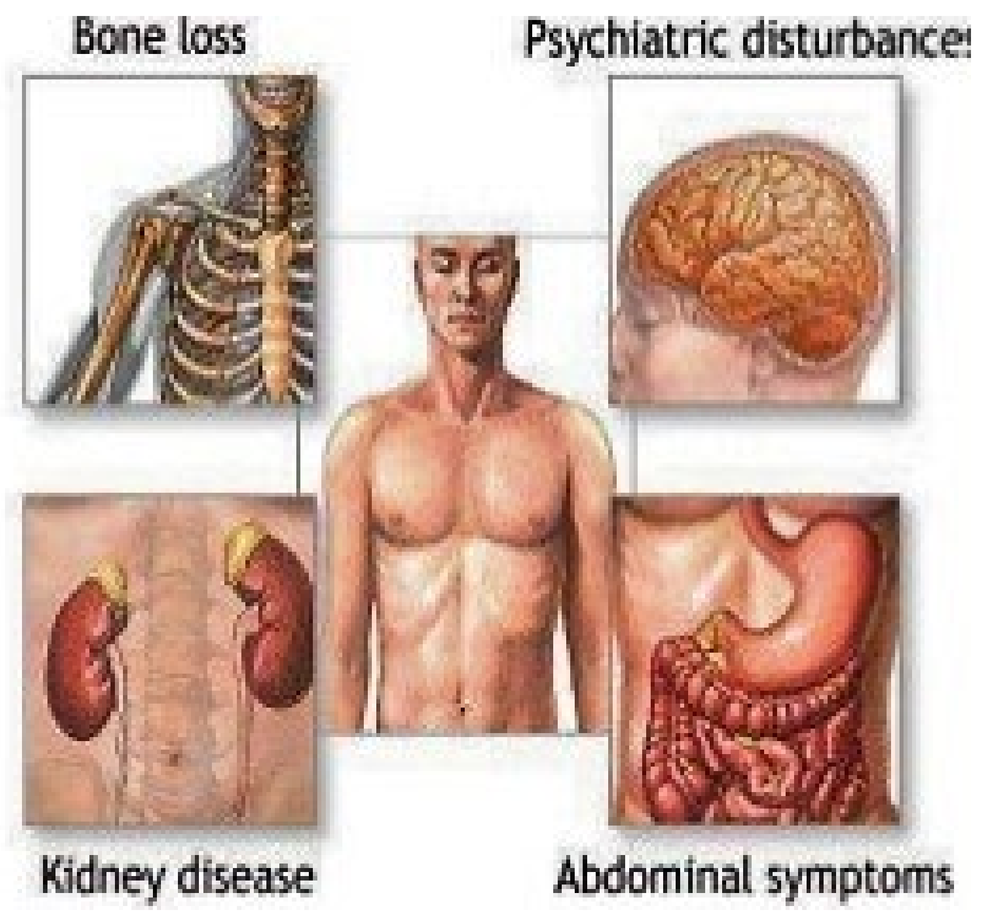
Solitary parathyroid adenoma (~85%)
Hyperplasia (~10-15%)
Double adenomas (1 – 2%)
Parathyroid carcinoma (~1%)



http://www.parathyroid.com/age.htm

Symptoms

- Classical symptoms: ‘bones, stones, abdominal groans & psychiatric overtones’ ARE RARE
- BONES: osteoporosis, fracture
- STONES: renal stones
- Abdominal GROANS: anorexia, nausea, constipation
- Psychiatric OVERTONES: fatigue, weakness, depressed mood, psychosis



Clinical Presentation

- Incidental finding of mild hypercalcaemia (Common presentation nowadays)
- Symptoms of severe hypercalcaemia:

  Weakness, tiredness, anorexia

  Nausea, vomiting, constipation, thirst, dry mouth and polyuria

  Mental confusion and drowsiness
- Renal complications:

  Renal stone and nephrocalcinosis, hypertension, renal failure
- Bone complications:

  Osteoporosis, bone pain, fractures
- Gastrointestinal manifestations: Epigastric pain / dyspepsia
- Joint pain (calcification of cartilage)
- Multiple endocrine neoplasia

PTH Action on Bone

- Continuously high level (e.g. Primary Hyperparathyroidism):

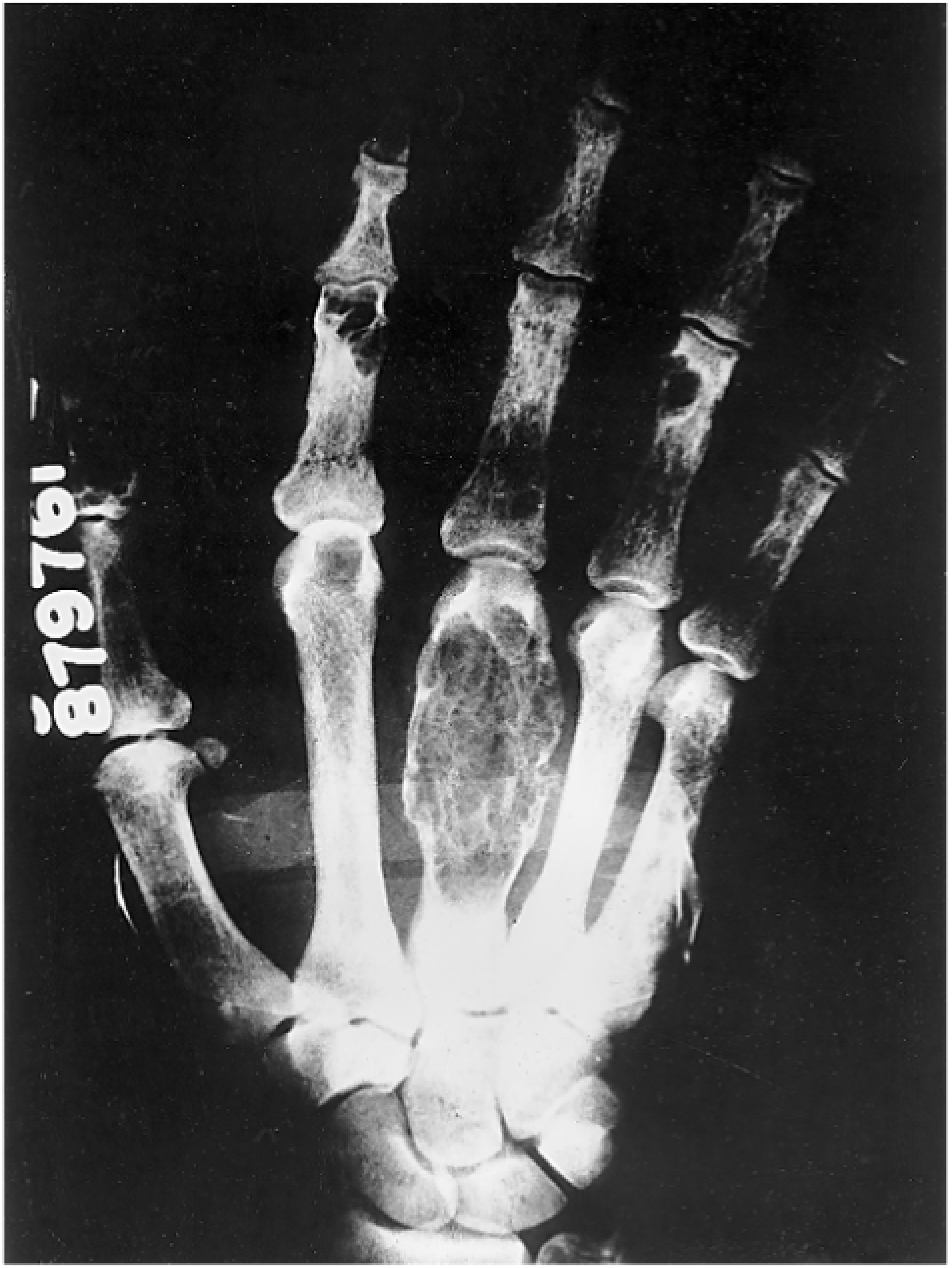
  Bone resorption (with cortical bone worse than trabecular bone)

- Intermittent low dose (e.g. Teriparatide):

  Anabolic action with increase in BMD

Bone Disorders in Hyperparathyroidism

- Demineralization with weakening of bones and even fracture
- Parathyroid bone disease
  - Cystic osteitis fibrosa and osteopenia
  - Bone pain, subperiosteal resorption
  - Bone deformities
- Severe bone disease is now uncommonly seen
- Low BMD especially in cortical bone more common (e.g. distal 1/3 of forearm)



- Osteoclastic resorption of smaller trabeculae
- Subperiosteal resorption
  - Most evident on phalanges
- Bone cysts
  - Mostly in central medullary portions of shafts of MCP, ribs or pelvis
- Osteoclastomas or brown tumours which
  - made up of numerous multinucleated
  - osteoclasts with stromal cells and matrix in
  - trabecular portions of the jaw, long bones, and ribs
- Skull salt and pepper appearance
- Pathologic fractures

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Presentation of PHPTH

	1930–1970	1970–2000
Nephrolithiasis	51–57%	17–37%
Hypercalciuria	36%	40%
Overt skeletal disease	10–23%	1·4–14%
Asymptomatic	0·6–18%	22–80%

Modified from reference 12.

Table 1: Changing clinical presentation of primary hyperparathyroidism

WD Fraser. Lancet 2009;374:145-158

Recommendations for Evaluation Asymptomatic PHPTH

- Biochemistry panel
  - Calcium, Phosphate, Alkaline Phosphatase (ALP)
  - Urea, Creatinine, eGFR
  - 25OHVitD
    - The disease is more active when subjects are vitamin D insufficient (<20ng/ml) or frankly deficient (<10ng/ml) (as defined by Institute of Medicine)
    - Reductions in PTH levels can occur when insufficient levels of 25(OH)D in PHPT are corrected
- PTH by second or third-generation immunoassay
- DXA for BMD (lumbar spine / hip / distal one third radius)
  - T score <-2.5 or less if peri or postmenopausal women AND men ≥ 50yo
  - Z score <-2.5 or less for premenopausal women and men < 50 yo
- Vertebral spine assessment (Xray or Vertebral Fracture Assessment by DXA)
- 24 hour urine for calcium and creatinine for fractional excretion of calcium
- Abdominal imaging by KUB, USG, CT scan for urinary stones
- Optional:
  - High resolution peripheral quantitative computed tomography (HRpQCT)
  - Trabecular bone score (TBS) by DXA
  - Bone turnover markers

JCEM 2018

Indications of Surgery in Asymptomatic PHPTH

Table 2. Guidelines for Surgery in Asymptomatic Primary Hyperparathyroidism: A Comparison of Current Recommendations with Previous Ones

Parameter	1990	2002	2008	2013	2022
Serum Calcium (>upper limit of normal)	1–1.6 mg/dL (0.25–0.4 mmol/L)	1.0 mg/dL (0.25 mmol/L)	1.0 mg/dL (0.25 mmol/L)	1.0 mg/dL (0.25 mmol/L)	1.0 mg/dL (0.25 mmol/L)
Skeletal	BMD by DXA: *Z*-score < −2.0 (site unspecified)	BMD by DXA: *T*-score < −2.5 at any site	BMD by DXA: *T*-score < −2.5 at any site Previous fragility fracture	a. BMD by DXA: *T*-score < −2.5 at lumbar spine, total hip, femoral neck or distal 1/3 radius b. Vertebral fracture by X-ray, CT, MRI, or VFA	a. BMD by DXA: *T*-score < −2.5 at lumbar spine, total hip, femoral neck or distal 1/3 radius* b.Vertebral fracture by X-ray, CT, MRI or VFA
Renal	a. eGFR reduced by >30% from expected. b. 24-Hour urine for calcium >400 mg/day (>10 mmol/day)	a. eGFR reduced by >30% from expected b. 24-Hour urine for calcium >400 mg/day (>10 mmol/day)	a. eGFR <60 cc/min b. 24-Hour urine for calcium not recommended	a. eGFR <60 cc/min b. 24-hour urine for calcium >400 mg/day (>10 mmol/day) and increased stone risk by biochemical stone risk analysis c. Presence of nephrolithiasis or nephrocalcinosis by X-ray, ultrasound, or CT	a. eGFR <60 cc/min** b. Complete 24-hour urine for calcium >250 mg/day in women (>6.25 mmol/day) or > 300 mg/day in men (>7.5 mmol/day) c. Presence of nephrolithiasis or nephrocalcinosis by X-ray, ultrasound, or CT
Age	<50 years	<50 years	<50 years	<50 years	<50 years

JBMR 2022

Table 2. Guidelines for Monitoring Patients with Asymptomatic PHPT Who Do Not Undergo Parathyroid Surgery: A Comparison of Current Recommendations With Previous Ones[a]

Serum calcium	Annually
Skeletal	Every 1–2 y (3 sites),[a] x-ray or VFA of spine if clinically indicated (eg, height loss, back pain)
Renal	eGFR, annually; serum creatinine, annually. If renal stones suspected, 24-h biochemical stone profile, renal imaging by x-ray, ultrasound, or CT

Pre-op Preparation

Preoperative Localization

- Guide surgeon planning the surgical strategy, NOT diagnostic procedure
- Negative or discordant imaging studies should not inhibit referral to an experience parathyroid surgeon

JCEM Oct 2014, 99:3595-3606

Preoperative Localization

- Parathyroid Ultrasound

  - Pre-operative or intra-operative

  - Limited in its ability to evaluate retroesphageal lesions or mediastinal parathyroid glands

- Nuclear Scintigraphy

  - ^{99m}Tc-Sestamibi Scan / Single photon emission CT (SPECT)

  - Principle: ^{99m}Tc-Sestamibi is taken up by the mitochondria in thyroid and parathyroid tissues; the radio tracer is only retained by the mitochondria-rich oxyphil cells in parathyroid glands and is longer than in thyroid tissue

  - Planar images obtained shortly after injection and again at approximate 2 hours to identify foci of retained radio-tracer activity consistent with a hyper-functioning parathyroid tissue

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Sestamibi Scan

Preoperative Localization

For negative localizations by USG and Sestamibi,

- Computed tomography
  - 4DCT (Multiple scans obtained after administration of IV contrast scans)
  - Parathyroid adenomas had rapid contrast uptake and washout
  - High radiation exposure
- Magnetic resonance imaging
- 18F-fluorocholine PETCT

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Surgery

Minimally Invasive Approach

- Subjected to multiple interpretations
- Commonly imply image-guided, focused operation
- Under local or regional anaesthesia
- Small incisions
- Open technique or endoscopic approaches
- Curative results can exceed 98% and nerve injury rates of <1%

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Minimally Invasive Approach

Potential advantages

- Improves cosmesis
- Shortens hospital stay
- Decreases wound pain
- Reduces morbidity
- Decreases overall cost

Pre-requisites

- Preoperative or intraoperative localization
- Intraoperative PTH Assay

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Intraoperative PTH Assay Monitoring

- To monitor success of parathyroid surgery
- Half-life of PTH:

  3.5 - 4 mins in patients with normal renal function
- Drop in serum PTH: 5 - 10 mins
- Drop in calcium: 24 - 48 hours
- 7 mins incubation and 15 minutes assay result
- Miami Criterion: Fall of 50% compared to the highest of either the pre-manipulation or pre-excision sample

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Parathyroid surgery for Multi-Glandular Disease (eg. in MEN1)

- Subtotal parathyroidectomy
  - Resection of 3 ½ glands
  - Preserve 50-80 mg vascularized gland
- Total parathyroidectomy

Total parathyroidectomy with immediate autotransplantation to forearm

Cryopreservation for possible delayed autotransplantation

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Beyond
Primary Hyperparathyroidism

Secondary and Tertiary Hyperparathyroidism

- Physiological response of PTH secretion to chronic hypocalcaemic state (e.g. chronic renal failure, vitamin D deficiency etc.)
- Prolonged secondary hyperparathyroidism may result in parathyroid hyperplasia and autonomous secretion of PTH, leading to tertiary hyperparathyroidism and hypercalcaemia

Medical Management of Primary Hyperparathyroidism

Calcimimetics

Mimics the action of calcium on tissues by allosteric action of the calcium-sensing receptor that is expressed in various human organ tissues

Cinacalcet in Primary Hyperparathyroidism

- Approved in European Medicines Agency (2008) and US Food and Drug Administration (2011)
- Specific indications in patients with primary hyperparathyroidism
- EMA: For patients in whom parathyroidectomy indicated but in whom surgery NOT clinically appropriate or is contraindicated
- FDA: For severe hypercalcaemia in patients unable to undergo parathyroidectomy
- Effective in lowering and often normalizing serum calcium
- Effects on intact serum PTH less pronounced
- No consistent observed effects on BMD

JCEM Oct 2014, 99:3607-3618

Treatment of osteoporosis in PHPT

Table 3. Trials Included in Systematic Review of Medical therapy[8]

Intervention	[Calcium] (mg/dL)	[PTH] (pg/mL)	Urinary Ca (mg/day)	Conclusions of systematic reviews (effect on bone density*)	Conclusions of systematic reviews (effect on serum calcium)	References
Alendronate	11.0 ± 0.5	170 ± 95	204 ± 109	Increases bone mineral density**	No effect	Chow and colleagues[193]; Khan and colleagues[190]; Rossini and colleagues[194]
Estrogen (raloxifene data are sparse)	10.6 ± 0.16	149 ± 32	240 ± 41	Increases bone mineral density***	Reduced***	Grey and colleagues[197]; Rubin and colleagues[198]
Cinacalcet	11.2 ± 0.45	138 ± 46	290.0 ± 120	No effect	Reduced**	Khan and colleagues[202]; Peacock and colleagues[201]
Denosumab	10.9 ± 0.1	115 ± 16	NR	Increases bone mineral density	No effect	Leere and colleagues[205]
Vitamin D	11.0 ± 0.3	83 ± 15	376 (320–432)	Increases bone mineral** density***	No effect	Lind and colleagues[184]; Roglighed and colleagues[182,183]

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Hypercalcaemia of Malignancy

Tumours most commonly linked with hypercalcaemia

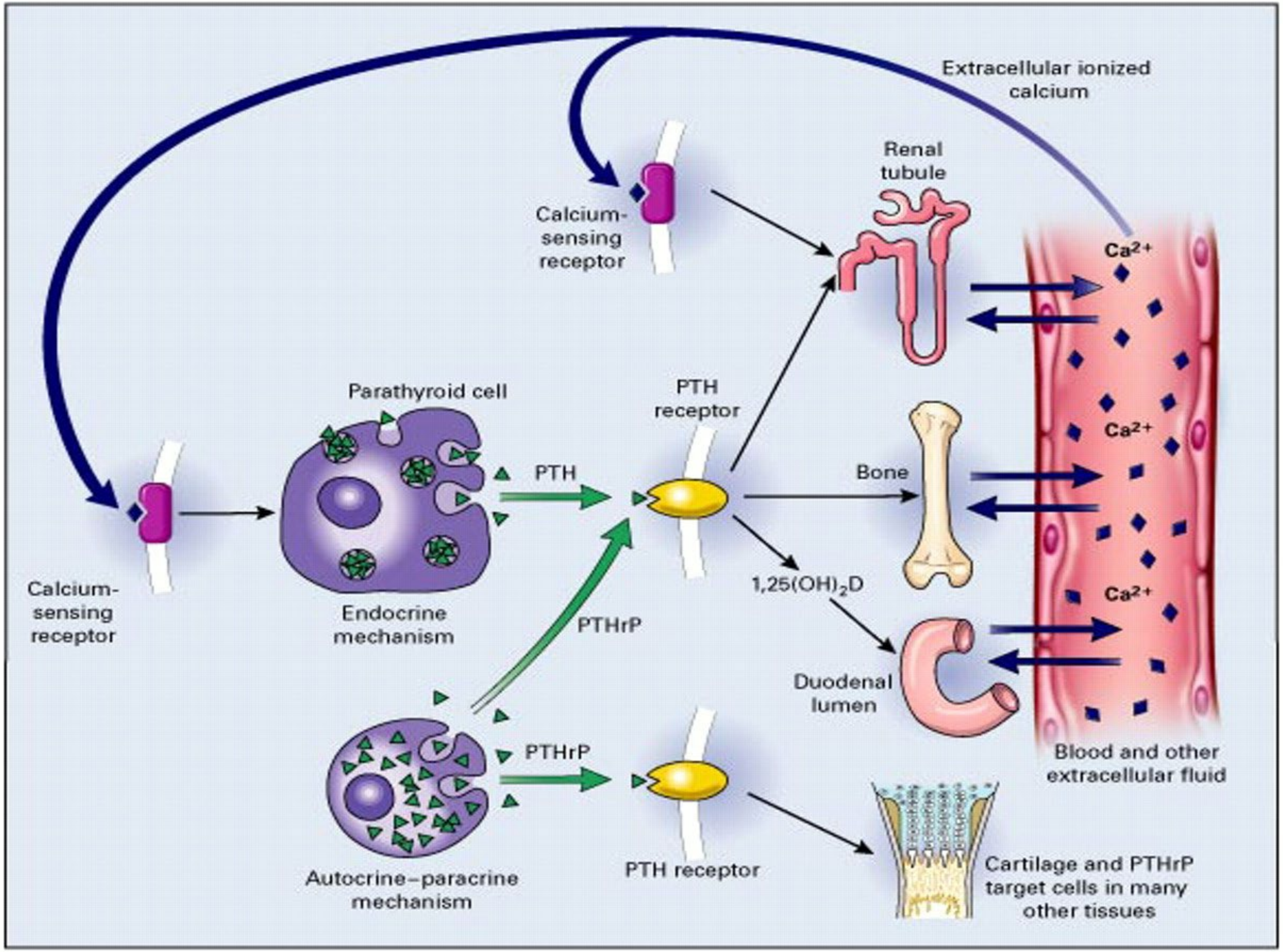
- Lung (squamous cell)
- Breast
- Head & neck (squamous)
- Kidney
- Myeloma
- Lymphoma

Mechanisms

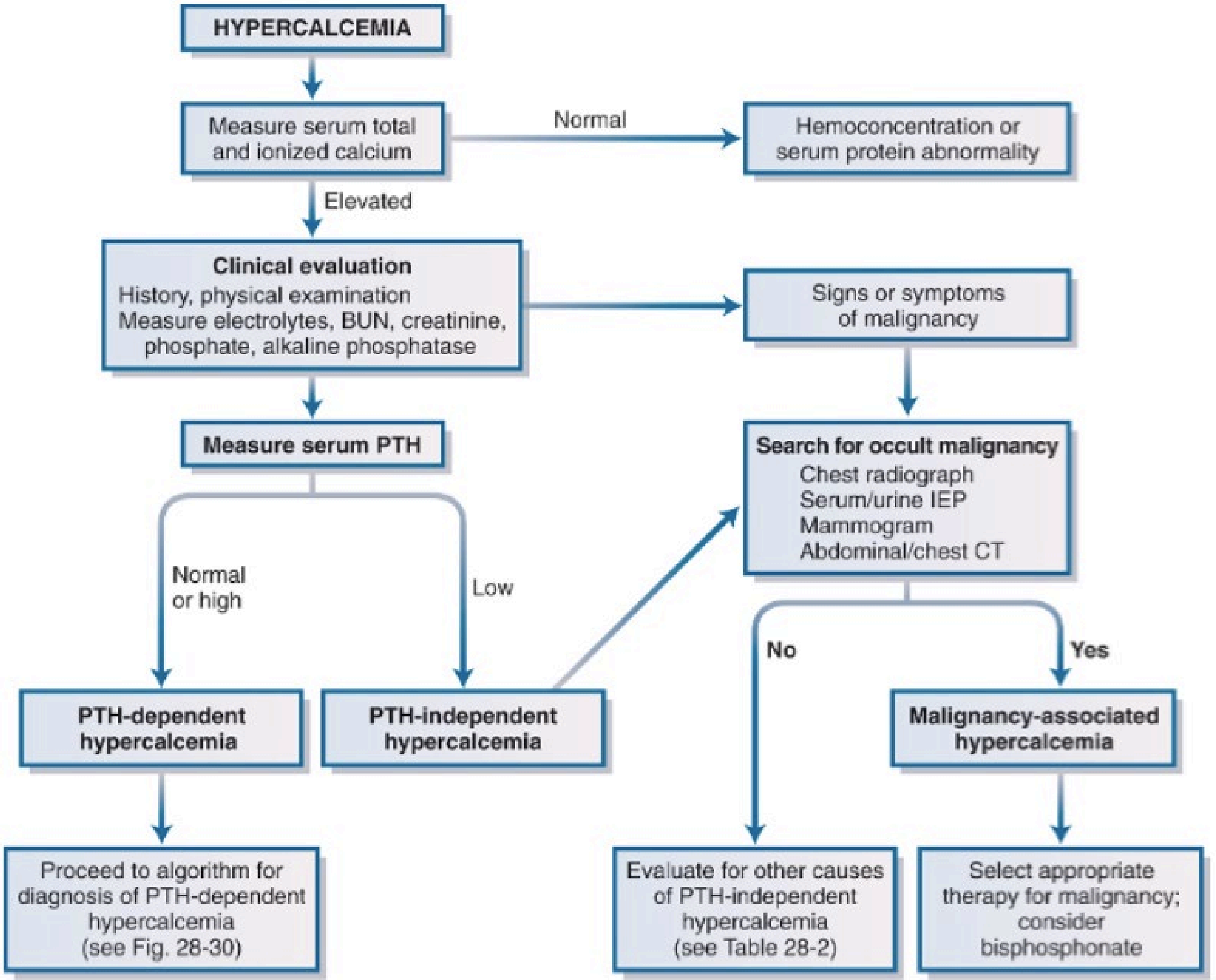
- Humoral mediated hypercalcaemia (PTH-related peptide; PTHrP)

e.g. non-metastatic solid tumours (SCC lung and H&N), renal cell carcinoma , bladder, breast, ovarian carcinomas

- Bone metastases with direct local destruction or induction of local osteolysis by tumour cells
- Lymphokine production by haematological malignancies that activate osteoclasts (e.g. multiple myeloma)
- Extra-renal production of 1,25 $(OH)_2$ VitD (e.g. adult T-cell lymphoma)



Extracellular ionized calcium
Renal tubule
Calcium-sensing receptor
Ca^{2+}
Parathyroid cell
PTH receptor
PTH
Bone
Ca^{2+}
Calcium-sensing receptor
Endocrine mechanism
$1,25(OH)_2D$
Ca^{2+}
PTHrP
Duodenal lumen
PTHrP
Blood and other extracellular fluid
Autocrine–paracrine mechanism
PTH receptor
Cartilage and PTHrP target cells in many other tissues



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Case Vignette

- Madam Chan
- F/52
- History of total thyroidectomy a few weeks ago
- Bilateral hand and peri-oral numbness
- Normal kidney function
- Blood tests showed low serum calcium, high phosphate with serum parathyroid hormone level at low normal range

Hypocalcaemia

Hypocalcaemia

- Symptoms of hypocalcaemia typically develop when adjusted serum calcium levels fall below 1.9 mmol/L, although this threshold varies
- Also depends on the rate of fall of serum calcium

Etiology

Most important: To confirm genuine hypocalcaemia! Look at serum albumin levels; measured ionised calcium

- Vitamin D deficiency (diet; malabsorption; chronic liver/renal disease)
- Hypoparathyroidism (idiopathic, familial, post-surgical)
- Magnesium deficiency
- Cytotoxic drug-induced hypocalcaemia (e.g Cisplatin)
- Pancreatitis
- Rhabdomyolysis
- Large volume blood transfusion

- Rare causes
  Resistance to PTH hormone
  (pseudohypoparathyroidism: post-receptor defect)
  Abnormal vitamin D synthetic pathway
  [1a hydroxylase deficiency or 1,25 $(OH)_2D$ resistance]

Turner J et al. Endocr Connect 2016

Clinical Presentation of Hypocalcaemia

- Perioral and digital paresthesiae
- Neuromuscular irritability (Trousseau's sign, Chvostek's sign)
  - Tetany and carpopedal spasm
- Convulsion
- Laryngeal spasm
- ECG: prolonged QT interval

Turner J et al. Endocr Connect 2016

Patient Scenario

Urea			4.8	4.1		2.8 - [illegible]
Creatinine			68	76		49 - 8[illegible]
Estimated GFR			>90	83 L		>90
F Glucose		5.8 H				See below
Calcium			1.95 L	2.14	1.69 L	2.11 - 2.[illegible]
Adjusted Calcium			1.89 L	2.06 L	1.57 L	2.11 - 2.[illegible]
Phosphate			1.60 H	1.60 H	1.59 H	0.88 - 1.45
Total Protein			70			67 - 87
Albumin			45	46	48	39 - 50
Globulin			25 L			26 - 40
Total Bili			5			4 - 23
ALP			59	54		32 - 93
ALT			12			7 - 36
AST			14			14 - 30
Urine Volume	1.32					
Duration	24					
Ur Creatinine	18922					5600 - 22000
Ur Calcium	9.6 H					2.0 - 7.4
Ur Calcium	0.51					See Below

Comment:

19CA093772 Result Telephoned.

Treatment of Hypocalcaemia

<u>Severe or Symptomatic Hypocalcaemia (Adjusted serum calcium <1.9 mmol/L)</u>

- ECG and Cardiac monitoring
- IV calcium gluconate by infusion

(e.g. 10% calcium gluconate 20ml IV over 10-15 minutes then 30ml 10% calcium gluconate in 500ml NS/D5 Q4-6H/pint)

- Monitor serum calcium level closely (e.g. q6-8h)

<u>Mild hypocalcaemia (Adjusted serum calcium >1.9 mmol/L)</u>

- Oral replacement of calcium
    - Caltrate = 600mg elemental calcium per tab
    - Oscal = 250mg elemental calcium per tab
    - Calcium gluconate = 27mg elemental calcium per tab
- Vitamin D (consider adding if no response after 2-4g elemental calcium)

Most important: Look out for Causes and Treat Accordingly

To Recap,

1) Calcium homeostasis
2) Clinical presentation of hypercalcaemia
3) Approach to hypercalcaemia
   - Interpretation of biochemical results
   - Further investigations and work-up for possible causes
   - General management of severe hypercalcaemia
4) Overall management of primary hyperparathyroidism
5) Approach to hypocalcaemia

Common Clinical Scenarios

	Calcium	Phosphate	PTH
Primary hyperparathyroidism	↑	↓ / ↔	↑ / ↔
Primary hypoparathyroidism	↓	↑ / ↔	↓ / ↔
Secondary hyperparathyroidism			
Chronic Renal Failure	↓ / ↔	↑	↔ /↑
Vitamin D insufficiency	↓ / ↔	↓ / ↔	↑
Tertiary hyperparathyroidism			
Vitamin D intoxication	↑	↑	↓
Bone Mets	↔ / ↑	↔	↔ / ↓

Thank you

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